Developer Setup Guide

This is a simple guide to set up the repositories for the CSV processor application code, infrastructure configs and manifests.

Repository Structure

Single monorepo — one clone gives you everything:

```
git clone https://github.com/asif-ahmedb/csv-processor
cd csv-processor
```

```
csv-processor/                        ← repo root (you are here after clone)
■■■ .github/workflows/ci.yml          ← CI — triggers on csv-processor-app/** changes
■■■ csv-processor-app/                ← Flask app, Docker image, tests
■■■ csv-processor-infrastructure/     ← AWS cluster (kops or Terraform/EKS) + S3
■■■ csv-processor-k8s-assets/         ← Helm chart, Ansible, Minikube scripts
```

Prerequisites

All Paths

Tool	Version
Git	2.x
Docker Desktop	4.x

Local / Minikube

Tool	Notes
Minikube	Local Kubernetes cluster
kubectl	Kubernetes CLI
Helm 3	Package manager for Kubernetes
Python 3.10+	For running tests locally

AWS

Tool	Notes
AWS CLI v2	Cloud provider CLI
kubectl	Kubernetes CLI
Helm 3	Package manager for Kubernetes
Git Bash	Required for `envsubst` on Windows
kops	Self-managed Kubernetes (kops deployments)
Terraform 1.5+	Managed EKS (Terraform deployments)

AWS IAM Permissions

The AWS user or role needs permissions to create and manage:

Required services: EC2, VPC, ELB, S3, IAM, Route 53 (kops), EKS (Terraform)

Verify credentials before starting:

```
aws sts get-caller-identity
```

Path 1 — Local Deployment

Run with Docker

```
cd csv-processor-app
docker build -t csv-processor:local .
docker run --rm -p 8081:8081 \
 -e S3_BUCKET="" \
 -e AWS_REGION=us-east-1 \
 csv-processor:local
```

Open **http://localhost:8081** and upload a CSV file.

Run Tests

```
cd csv-processor-app/web
pip install -r requirements.txt
pytest tests/ -v
```

Deploy on Minikube

```
cd csv-processor-k8s-assets/minikube
./deploy.sh
kubectl port-forward -n csv-processor svc/csv-processor 8080:80
```

Open **http://localhost:8080**

Path 2 — AWS with kops

Configure

```
cd csv-processor-infrastructure/kops
cp settings.env.example settings.env
```

Edit `settings.env` and set:

```
export AWS_ACCOUNT_ID=<YOUR_ACCOUNT_ID>
export AWS_REGION=us-east-1
```

Bootstrap Cluster (~15 min)

```
chmod +x scripts/*.sh
./scripts/bootstrap.sh
```

Deploy Application

```
cd ../../csv-processor-k8s-assets
source ../csv-processor-infrastructure/kops/settings.env
helm upgrade --install csv-processor ./helm/csv-processor \
 -f ./helm/csv-processor/values-kops.yaml \
 --namespace csv-processor --create-namespace \
 --set config.s3Bucket="${PROCESSED_BUCKET}" \
 --set-string aws.accountId="${AWS_ACCOUNT_ID}"
```

Access

```
kubectl get svc -n csv-processor csv-processor -w
# Open http://<EXTERNAL-IP> once available
```

Teardown

```
cd csv-processor-infrastructure/kops
./scripts/teardown.sh --yes
```

Path 3 — AWS with Terraform / EKS

Provision Infrastructure

```
cd csv-processor-infrastructure/terraform
cp terraform.tfvars.example terraform.tfvars
terraform init
terraform plan
terraform apply
```

Configure kubectl

```
aws eks update-kubeconfig \
 --name $(terraform output -raw eks_cluster_name) \
 --region $(terraform output -raw aws_region)
```

Deploy Application

```
cd ../../csv-processor-k8s-assets
helm upgrade --install csv-processor ./helm/csv-processor \
 -f ./helm/csv-processor/values-eks.yaml \
 --namespace csv-processor --create-namespace \
 --set config.s3Bucket=$(terraform -chdir=../csv-processor-infrastructure/terraform output -raw s3_buck
 --set aws.irsaRoleArn=$(terraform -chdir=../csv-processor-infrastructure/terraform output -raw app_irs
```

Access

```
kubectl port-forward -n csv-processor svc/csv-processor 8080:80
# Open http://localhost:8080
```

Teardown

```
cd csv-processor-infrastructure/terraform
terraform destroy
```

CI/CD

GitHub Actions workflow triggers on every push to `main`:

1. Run **pytest**
2. Build **Docker image**
3. Push image to **Docker Hub**

Required Secrets

Add these in the **csv-processor** repository: **Settings** → **Secrets and variables** → **Actions**

Secret	Description
`DOCKERHUB_USERNAME`	Docker Hub username
`DOCKERHUB_TOKEN`	Docker Hub access token

Reference Documents

Guide	Location in monorepo
kops provisioning	csv-processor-infrastructure/kops/KOPS_QUICK_REFER
Application deployment	csv-processor-k8s-assets/DEPLOY_QUICK_REFERENC
Minikube	csv-processor-k8s-assets/minikube/MINIKUBE_QUICK_R